



DBD281 GROUP PROJECT

FIFA 2026 Tournament Database



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Database normalization

This project highlights a design and implements a relational database system for managing the 2026 FIFA World Cup tournament. The system supports teams, players, Stadiums, matches, fans, countries, coaching staff, player match stats and ticket bookings. The objective is to ensure data integrity, efficient data storage and data retrieval using Microsoft SQL server and T-SQL.

0NF:

- Unstructured data with repeating groups

MatchID	Team	Players(multiple)	PlayerStats(multiple)	Stadium	County	Fans(multiple)	Tickets(multiple)
1	Team A	Player1, player2	Goals, Assists	StadiumX	StadiumY	Fan1, Fan2	Ticket1, ticket2

- No repeating groups
- Hard to Query
- Data duplication
- unorganised/messy

1NF

Rule:

- Remove repeating groups thus removing redundancy
- Atomic values only

Matches	MatchID(PK)	HomeTeamID	AwayTeamID	StadiumID	Stage	Homescore	Awayscore
Teams	TeamID(PK)	CountryID	TeamName	CoachName	Ranking		
Players	PlayerID(PK)	TeamID	FirstName	Lastname	DOB	Jerseynumber	
PlayerMatchStats	MatchID(PK,FK)	PlayerID	Goals	Assists	Yellowcards	RedCards	
Tickets	FanID(PK)	CountryID	FirstName	Lastname	email	phone	
Fans	StadiumID(PK)	CountryID	StadiumName	City	Capacity		
Countries	CountryID (PK)	CountryID	FirstName	LastName	Email	Phone	
CoachingStaff	StaffID(PK)	TeamID	Firstname	LastName	Role		

2NF

Rule:

- Remove partial dependencies(depending on the primary key)

Matches	MatchID(PK)	HomeTeamID	AwayTeamID	StadiumID	Stage	Homescore	Awayscore
Teams	TeamID(PK)	CountryID	TeamName	CoachName	Ranking		
Players	PlayerID(PK)	TeamID	FirstName	Lastname	DOB	Jerseynumber	
PlayerMatchStats	MatchID(PK,FK)	PlayerID	Goals	Assists	Yellowcards	RedCards	
Tickets	FanID(PK)	CountryID	FirstName	Lastname	email	phone	
Fans	StadiumID(PK)	CountryID	StadiumName	City	Capacity		
Countries	CountryID (PK)	CountryID	FirstName	LastName	Email	Phone	
CoachingStaff	StaffID(PK)	TeamID	Firstname	LastName	Role		

3NF

Rule:

- Remove transitive dependencies

Matches	MatchID(PK)	HomeTeamID	AwayTeamID	StadiumID	Stage	Homescore	Awayscore
Teams	TeamID(PK)	CountryID(FK)	TeamName	Ranking			
Players	PlayerID(PK)	TeamID(FK)	FirstName	Lastname	DOB	Jerseynumber	
PlayerMatchStats	MatchID(PK,FK)	PlayerID(PK,FK)	Goals	Assists	Yellowcards	RedCards	
Tickets	TicketID(PK)	MatchID(FK)	FanID(FK)	PurchaseDate	TicketPrice	Seatnumber	
Stadium	StadiumID(PK)	CountryID(FK)	StadiumName	City	Capacity		
Countries	CountryID (PK)	CountryName	Continent				
Fans	FanID(PK)	CountryID(FK)	Firstname	LastName	Email	Phone	
CoachingStaff	StaffID(PK)	TeamID(FK)	Firstname	LastName	Role		

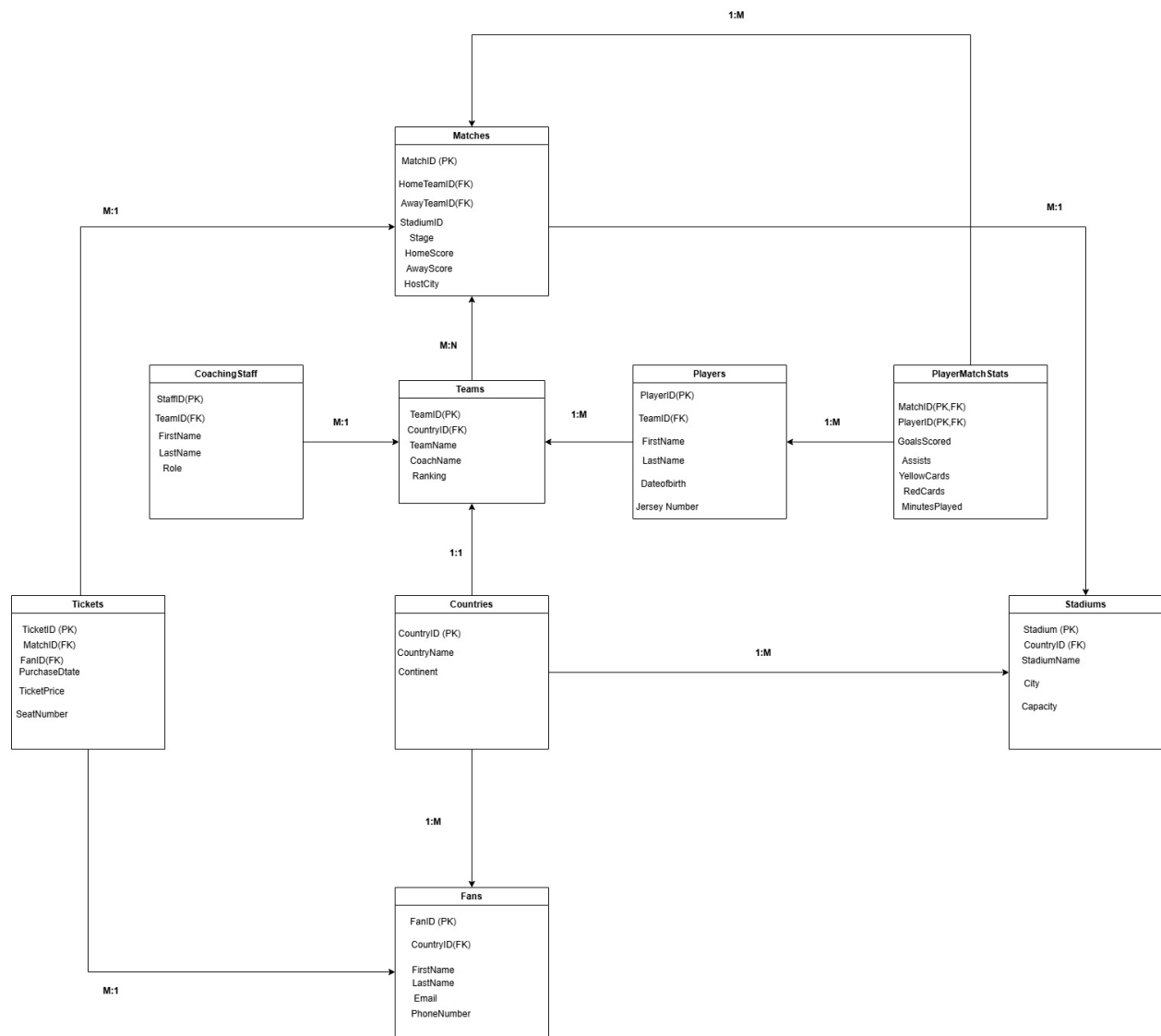
The database has been normalised to third normalized form ensuring minimal redundancy and optimal data integrity.

Business rules

- Each team must have a unique identifier like TeamID.
- A stadium can host many matches, but each match occurs in 1 stadium.
- A team can have multiple players but the condition is for each player to belong to each team.
- A match must involve exactly 2 teams.
- A ticket must be linked to a specific match.
- Each player must belong to a valid team for referential integrity.

7. A team cannot play against itself.
8. Each match must have a scheduled date and time.
9. Each booking must reference a valid fan and match.
10. The number of tickets sales cannot exceed stadium capacity.
11. Each ticket price must be greater than 0.
12. A fan can book multiple tickets but each booking must reference a valid match.
13. Each team must have 1 coach

ERD Diagram



ERD Table

Table Name	Attributes	Primary key	Foreign key
Countries	CountryID, CountryName, Continent"	CountryID	
Stadiums	StadiumID, CountryID, StadiumName, City, Capacity	StadiumID	CountryID
			→Countries
Teams	TeamID, CountryID, TeamName, CoachName, Ranking	TeamID	CountryID → Countries
CoachingStaff	"StaffID, TeamID, FirstName, LastName, Role	StaffID	TeamID → Teams
Players	"PlayerID, TeamID, FirstName, LastName, DateOfBirth, JerseyNumber	PlayerID	TeamID → Teams
Matches	"MatchID, HomeTeamID, AwayTeamID, StadiumID, Stage, HomeScore, AwayScore, HostCity ,MatchDate	MatchID	HomeTeamID → Teams
			AwayTeamID → Teams
			StadiumID → Stadiums
PlayerMatchStats	, MatchID, PlayerID, GoalsScored, Assists, YellowCards, RedCards ,MinutesPlayed	Match + PlayerID	MatchID → Matches
			PlayerID → Players
Fans	FanID, CountryID, FirstName, LastName, Email, PhoneNumber	FansID	CountryID → Countries
Tickets	TicketID, MatchID, FanID, PurchaseDate , SeatNumber, TicketPrice	TicketID	MatchID → Matche
			FanID → Fans